

Algorithms in C++: Programming Assignment 1

1. Objective

Your goal is to write a program that determines if a string has all unique uppercase letters.

2. Problem

The string will be read as a command line argument to the program. The program will then check if the string has all uppercase letters. If it does not, the program will display an error message and exit. Otherwise, the program will determine whether or not the string has all unique uppercase characters.

Case 1: No input arguments
Print usage message.

```
$ ./unique
Usage: ./unique <string>
```

Case 2: Too many input arguments
Print usage message.

```
$ ./unique too many words
Usage: ./unique <string>
```

Case 3: Bad input
\$./unique HI
Error: String must contain only uppercase letters.

Case 4: Bad input
\$./unique 1234567890
Error: String must contain only uppercase letters.

Case 5: Unique
\$./unique ABCDEFGHIJKLMNOPQRSTUVWXYZ
All letters are unique.

Case 6: Duplicates
\$./unique LONGSTRING
Duplicate letters found.

3. Hint

When a letter is seen, check if the corresponding bit is equal to 1. If it is, the letter has already been seen and must be a duplicate. Otherwise, change the corresponding bit from 0 to 1.